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# Lab 11 (solution)

FTP MachLe MSE  
FS 2026

## Unsupervised Learning: Clustering

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Machine Learning  
WÜRC

After this unit, ...

### Lernziele/Kompetenzen

- you know the *three* clustering algorithms: *k-means*, *dbscan* and *agglomerative clustering* (using average, complete or Ward linkage).
  - you are able to explain the working principle of *k-means*, *dbscan* and *agglomerative clustering*, their advantages and disadvantages and to apply them to data using **scikit-learn** in Python.
  - you are able to plot the **inertia** and to determine the elbow point of this curve to find an optimum number of clusters as *hyperparameter*.
  - you know the way how to *evaluate* a cluster algorithm using *metrics*, namely using **ARI** (*adjusted rand index*), **NMI** (*normalized mutual information*), **SC** (*silhouette score*) and *inertia*.
  - you are able to correctly *scale* the data before clustering is applied especially (MinMax, StandardScaler, RobustScaler) or to meaningfully **transform** the data (eg. using PCA, **t-SNE** or **NMF**) before a clustering algorithm is applied.
  - you know what a *Gaussian Mixture Model* **GMM** is and how the *expectation maximization algorithm* (EM algorithm) works. You are able to interpret the **kMeans** algorithm as a form of an EM algorithm with an E-step and an M-Step.
  - you are able to apply clustering on the faces dataset (agglomerative, k-means and dbscan) to detect and *group* similar faces.
  - you are able to apply a *hierarchical cluster analysis* on a voting dataset.
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### 1. Clustering Algorithms [M,I]

This clustering algorithm initially assumes that each data instance represents a single cluster.

| Welche der folgenden Aussagen sind wahr und welche falsch? | wahr | falsch |
|------------------------------------------------------------|------|--------|
| a) agglomerative clustering                                | ●    | ○      |
| b) t-SNE                                                   | ○    | ●      |
| c) k-means clustering                                      | ○    | ●      |
| d) expectation maximization                                | ○    | ●      |

### 2. Elbow Curve and `sklearn.cluster.KMeans` [A,I]

The solution Jupyter notebook can be found on moodle:

`Lab11_A2_MakeBlobs_kMeans.ipynb`

### 3. k-Means, Gaussian Mixture Models and the EM algorithm [A,II]

The solution Jupyter notebook can be found on moodle:

`Lab11_A3_EM_KMeans_MixtureModels.ipynb`

### 4. Image compression using `kMeans` [A,II]

The solution Jupyter notebook can be found on moodle:

`Lab11_A4_ImageCompressionUsingKMeans.ipynb`

### 5. Detecting similar faces using DBSCAN [A,II]

The solution Jupyter notebook can be found on moodle:

`Lab11_A5_DBSCAN_DetectSimilarFaces.ipynb`